

AI & Machine Learning Engineering Technical Competency Report

Specialization: Machine Learning Engineer / AI Developer
Focus Areas: Supervised Learning, Optimization, Signal Processing, AI Systems Engineering
Evaluation Date: July 2026

1. Programming Languages & Core Tech Stack

Language / Skill	Proficiency Level	Rating	Technical Capabilities & Application
Python	Advanced / Production-Ready	4.3 / 5.0	Data structures, object-oriented programming for ML, pipeline automation, NumPy vectorization, data manipulation with Pandas, and ML algorithm implementation.
C# / .NET	Advanced	4.0 / 5.0	Integrating ML artifacts into production backend services, API communication, strong OOP principles, and asynchronous processing.
SQL	Intermediate to Advanced	3.8 / 5.0	Data extraction, aggregating features for ML, query optimization, handling relational schemas for analytics.
Mathematics for AI	Solid Core Foundation	3.9 / 5.0	Linear Algebra, Multi-variable Calculus, Mathematical Optimization

Language / Skill	Proficiency Level	Rating	Technical Capabilities & Application
			(Gradient Descent, Cost Functions), Probability & Statistics for ML modeling.

2. Frameworks, Libraries & Tools Matrix

Tool / Library	Category	Proficiency	Use Cases & Demonstrated Mastery
Scikit-Learn	Machine Learning Framework	4.4 / 5.0	Classification, Regression, Preprocessing, Pipeline construction, Model Selection, GridSearchCV, Feature Importance, Class Weighting.
NumPy	Numerical Computing	4.2 / 5.0	Vectorized matrix calculations, custom ML algorithm construction from scratch (Logistic Regression, Decision Trees), mathematical optimization.
Pandas	Data Wrangling	4.3 / 5.0	Exploratory Data Analysis (EDA), Feature Selection, Data Cleaning, Handling missing values, GroupBy aggregations, Pivot Tables.
Matplotlib & Seaborn	Data Visualization	3.9 / 5.0	Visualization of evaluation metrics, Feature Importances,

Tool / Library	Category	Proficiency	Use Cases & Demonstrated Mastery
			Data Distributions, ROC-AUC curves, and Confusion Matrices.
Joblib	Serialization & MLOps	4.1 / 5.0	Model persistence, saving scalers and feature schemas, preparing models for deployment in external environments.
SpeechRecognition & DSP Libraries	Signal Processing / Audio AI	3.7 / 5.0	Audio noise reduction algorithms, speech-to-text pipeline creation, and audio data filtering for Digital Signal Processing.

3. Deep Dive into AI & ML Engineering Competencies

A. Machine Learning Algorithms Mastery

Sub-Topic	Technical Description & Application	Rating
Supervised Learning Algorithms	Hands-on implementation and fine-tuning of Logistic Regression, Decision Trees, and Random Forest Classifiers.	4.4 / 5.0
Algorithm Construction from Scratch	Deep mathematical understanding by constructing Logistic Regression pipelines using raw NumPy vectorization and optimization routines.	4.2 / 5.0
Handling Imbalanced Datasets	Application of cost-sensitive learning	4.4 / 5.0

Sub-Topic	Technical Description & Application	Rating
	(class_weight='balanced'), metric selection (Recall vs. Precision), and Stratified Cross-Validation.	

B. End-to-End Pipeline Engineering

Sub-Topic	Technical Description & Application	Rating
Feature Engineering & Selection	Conducting feature importance pruning experiments (e.g., Top 10 vs. Top 8) to eliminate noise and mitigate overfitting.	4.3 / 5.0
Data Preprocessing & Scaling	Preventing Data Leakage by fitting Scalers (e.g., StandardScaler) strictly on training subsets before transformation.	4.3 / 5.0
Automated Hyperparameter Tuning	Extensive experience with GridSearchCV, parameter grid design, and 5-Fold Cross-Validation.	4.4 / 5.0

C. Model Evaluation & Explainable AI (XAI)

Sub-Topic	Technical Description & Application	Rating
Evaluation Metrics Optimization	ROC-AUC Score optimization (achieved 0.9925), Recall Optimization (achieved 88.0% for minority class), Precision, F1-Score, and Confusion Matrix analysis.	4.5 / 5.0
Business Interpretability & XAI	Translating Tree Feature Importance into domain-specific actionable insights (e.g., Engagement &	4.4 / 5.0

Sub-Topic	Technical Description & Application	Rating
	Recency analysis).	

4. Applied AI Projects & Engineering Showcase

- **1. E-Commerce Customer Churn Prediction Engine:** Python, Scikit-Learn, Pandas, NumPy, Joblib, GridSearchCV. Optimized a Tuned Random Forest model to achieve **88% Recall** and **0.9925 ROC-AUC**.
- **2. Custom ML Algorithms Implementation (From Scratch):** Python, NumPy, Linear Algebra. Built Logistic Regression and optimization workflows directly without high-level ML wrappers.
- **3. Audio Signal Processing & Speech Recognition Pipeline:** Python, DSP Filters, SpeechRecognition. Noise removal preprocessing pipeline for raw audio integrated with speech-to-text.

5. Technical Assessment Summary

Demonstrates solid proficiency as an **Applied AI / ML Engineer** with an overall rating of **4.1 / 5.0**, possessing a strong functional mathematical foundation, robust pipeline engineering skills, and clean software practices.